LDLS Project 0

Yash Priya Shastri & Muffadal Alavi

Domain

LaTeX papers: *sections, theorem-like environments and display math*

Tree Statistics

document	nodes	tokens	blocks	depth	subtree tokens (min/mean/max)	pure markup
1_transformer_architecture	39	1324	10	3	54 / 143 / 379	5%
2_kv_cache_management	41	1481	11	3	51 / 143 / 259	5%
3_mathematical_results	28	842	7	3	54 / 124 / 276	20%
4_structured_documents	50	1639	14	3	46 / 125 / 271	2%
5_attention_masks	51	1729	14	3	21 / 122 / 292	5%
total		7015	56			

Depth counts levels below the document root. "Subtree tokens" is the size of each block that can be evicted, so the root (whose subtree is the whole document) is excluded. "Pure markup" is the share of tokens in headings and in nodes whose rendered text carries no prose.

CPU Run Times

document	prefill	baseline answer	replace + evict
1_transformer_architecture	174.6 s	187.5 s	413.1 s
2_kv_cache_management	200.6 s	213.9 s	350.3 s
3_mathematical_results	82.7 s	94.4 s	231.6 s
4_structured_documents	185.1 s	196.7 s	430.2 s
5_attention_masks	224.6 s	236.1 s	466.7 s

Model: Llama-3.2-1B-Instruct-Q4_K_M, CPU only, greedy decoding. Every column is wall clock for one full driver invocation, including model load, and all of them ran single threaded so that the logits stay reproducible. The isolation results do not depend on the thread count; the timings do.

Test Results

document	changed subtree	tokens	answer	max logit shift	KV before	KV after	freed
1_transformer_architecture	3	379	unchanged	0.0000	41.4 MiB	30.5 MiB	10.9 MiB
2_kv_cache_management	9	259	unchanged	0.0000	46.3 MiB	39.3 MiB	7.0 MiB
3_mathematical_results	4	276	unchanged	0.0000	26.3 MiB	18.7 MiB	7.7 MiB
4_structured_documents	9	271	unchanged	0.0000	51.2 MiB	43.7 MiB	7.5 MiB
5_attention_masks	9	292	unchanged	0.0000	54.0 MiB	45.9 MiB	8.1 MiB

Both checks were run per document: replace refilled the changed subtree with new text, evict removed it outright.

Positive control: evicting the subtree the answer comes from changed the answer (max logit shift 9.89), so the isolation results above are not vacuous.